

VAX (Video-Audio to “X”)

Using Existing Video and Audio-based Activity Recognition Models to Bootstrap Privacy-Sensitive Sensors



Praseon
Patidar



Mayank
Goel



Yuvraj
Agarwal

Emerging Sensing Modalities for Human Activity Recognition



DOPPLER RADAR
(MOVEMENT)



LIDAR
(POSITION)



INFRARED
(THERMAL)



However, these modalities need in-situ training data since:

- (a) No existing ML models, activities labels needed
- (b) ML models do not generalize in new placements or locations

HAR with Audio-Video Sensing is more Accessible



Availability of Large-Scale Datasets.



AVA [cite]



NTU120 [cite]



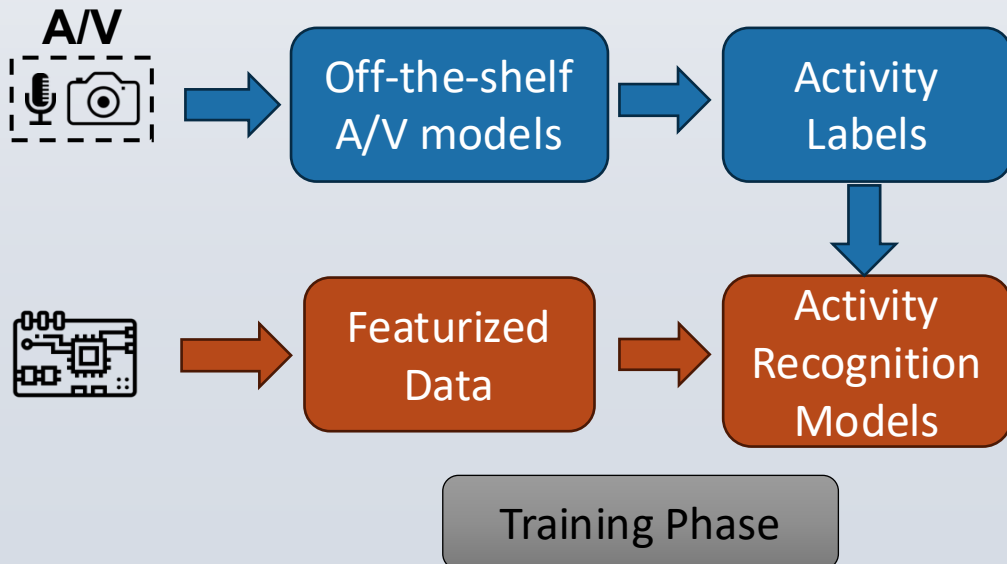
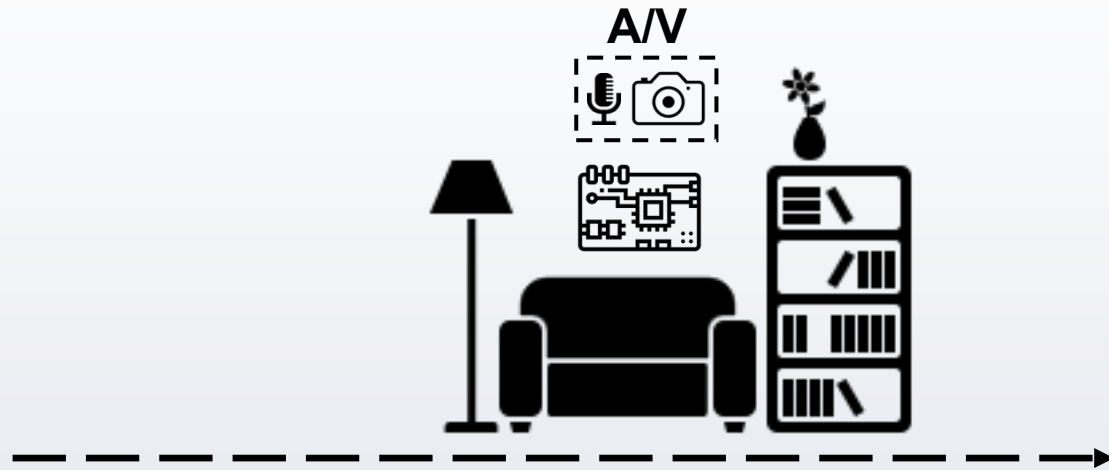
Availability of pre-trained models, video toolboxes.



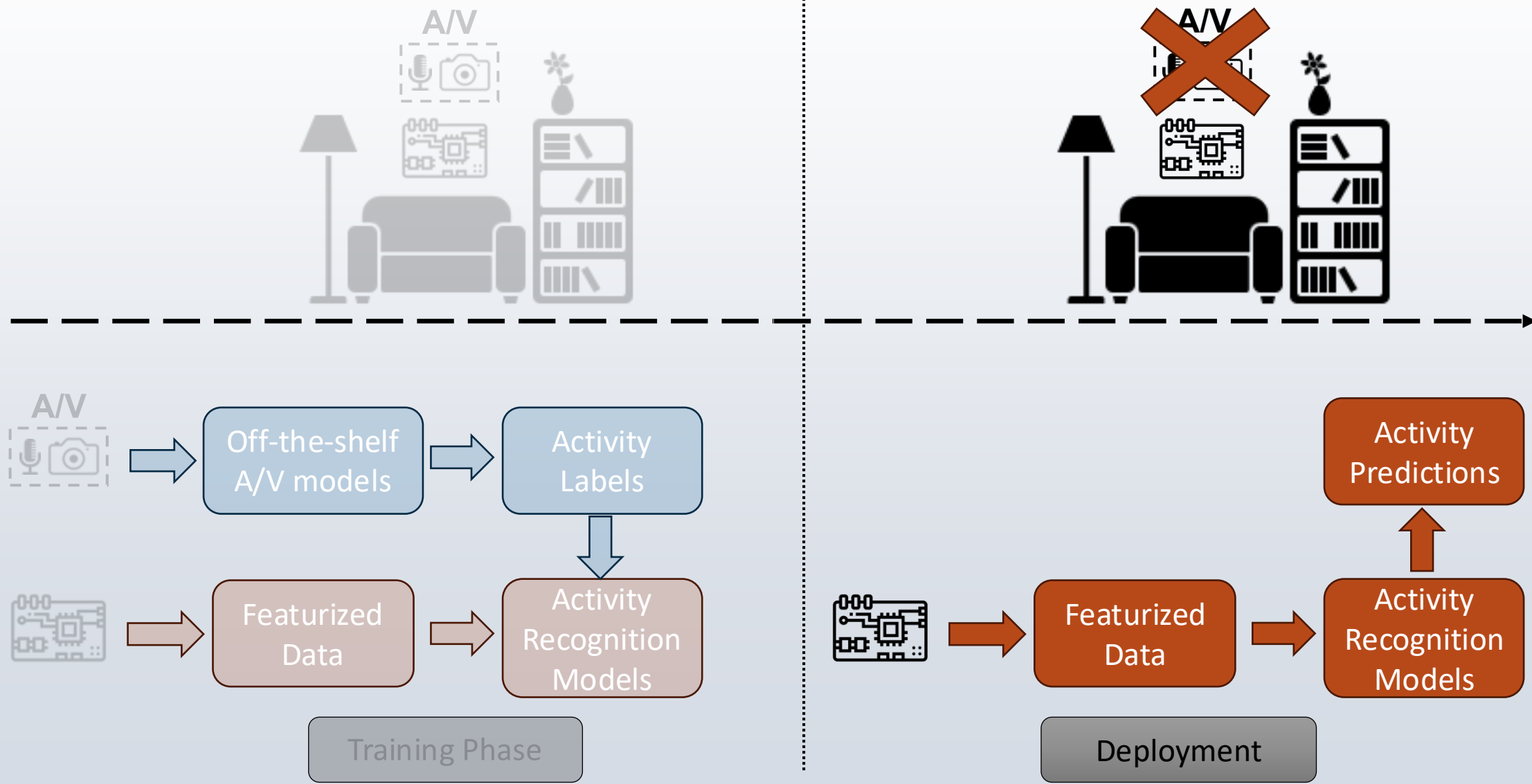


Using audio-video (A/V) models to bootstrap other (“X”) sensing modalities in new surroundings.

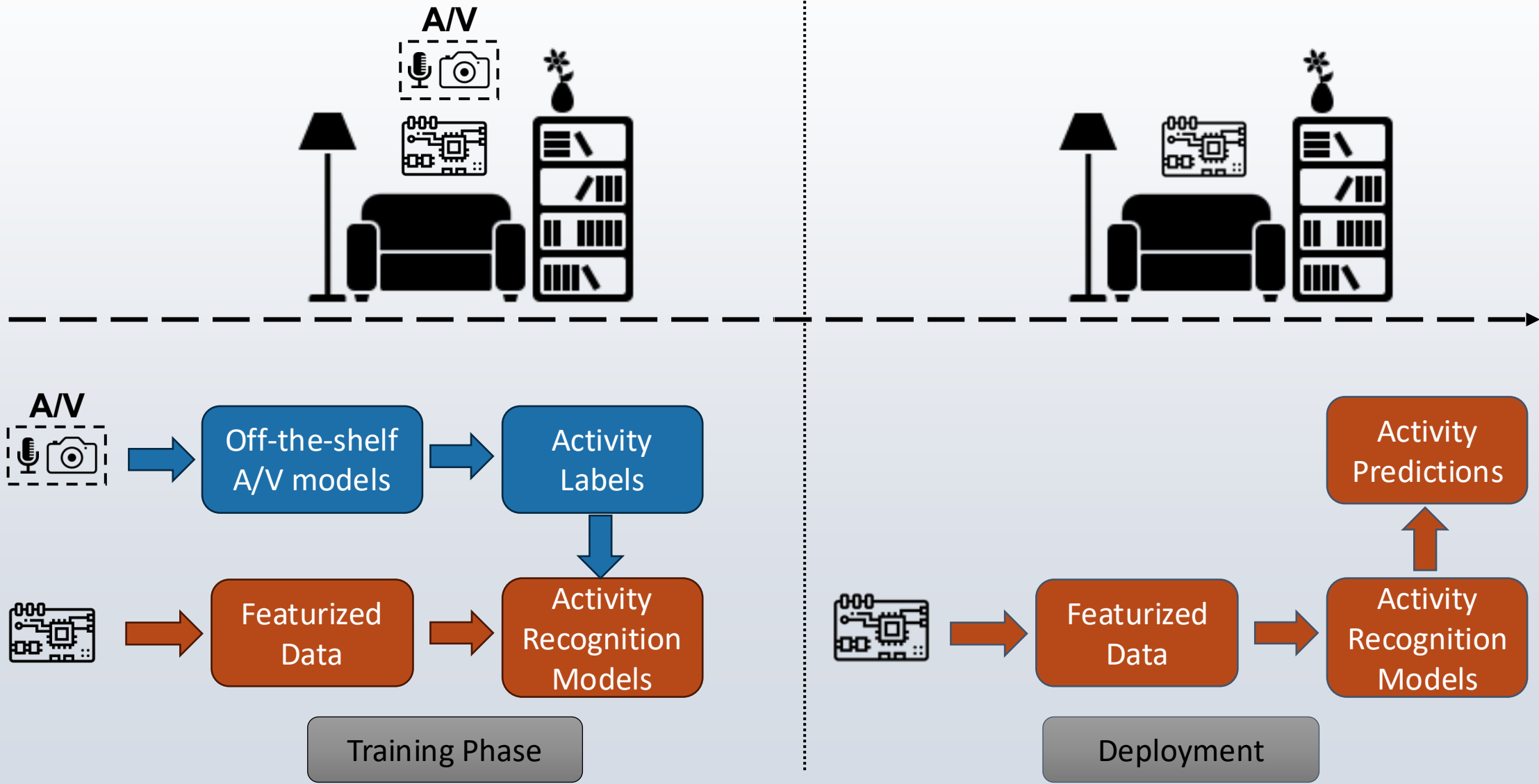
VAX (Video/Audio to “X”)



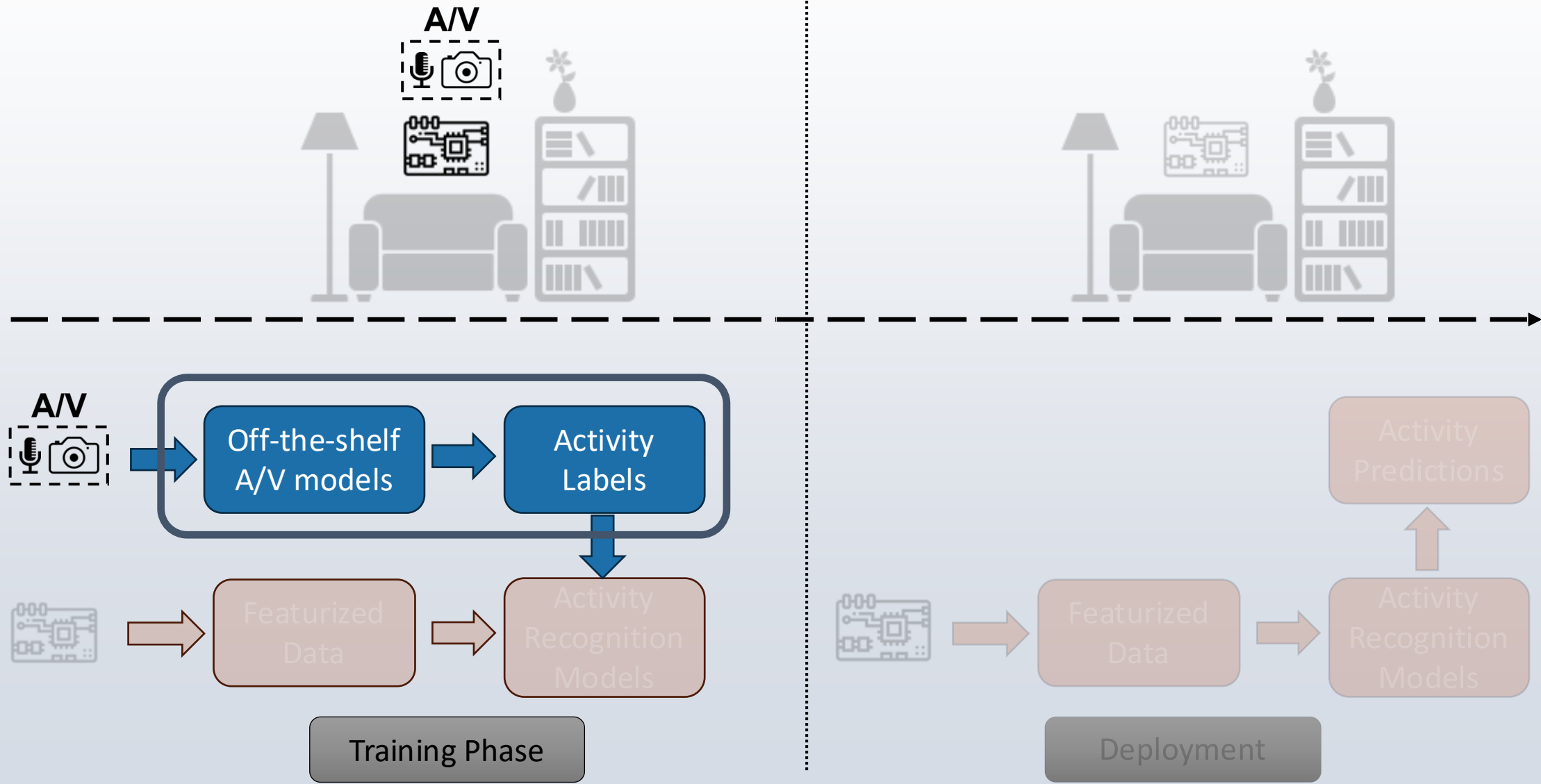
VAX (Video/Audio to “X”)



VAX (Video/Audio to “X”)



VAX (Video/Audio to “X”)

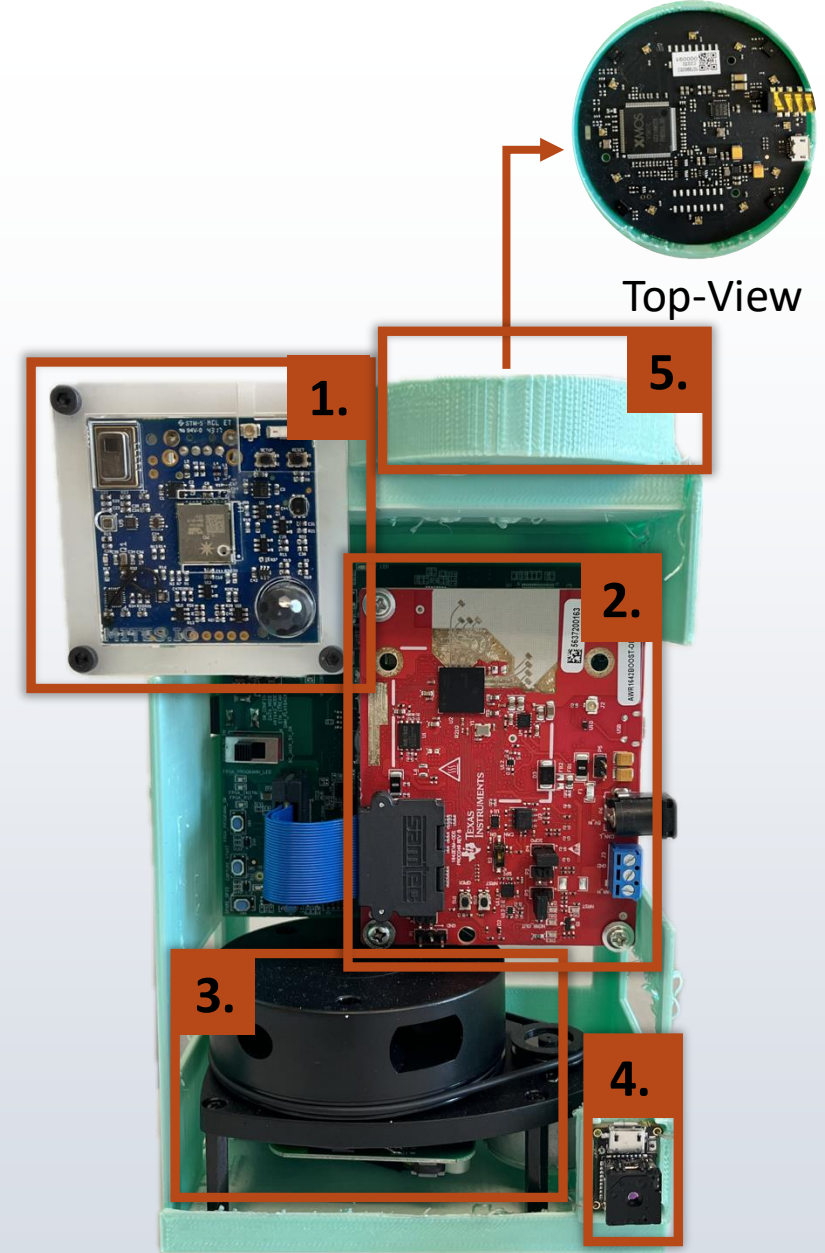


Hardware Setup

1. Mites Custom Device with 9 Sensors ^[1]
2. FMCW Doppler Radar - Movement Sensing
3. 2-D Lidar - Position Sensing
4. Thermal Array* - Infrared Sensing
5. Microphone Array - Sound Localization

*Denatured to 10x8 size to reduce privacy concerns

[1] Boovaraghavan et. al, 2023. Mites: Design and Deployment of a General-Purpose Sensing Infrastructure for Buildings. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 7(1), pp.1-32.



Data Collection

- # Homes: 10
- # Activities per home: 17
- # Instances per activity: 5
- # Total: 850

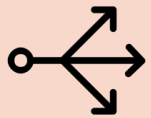


Kitchen: Chopping/Grating, Baking, Washing Dishes, Using Blender, Using Microwave, Using Fridge, Cooking on Stove

Bathroom: Using Hairdryer, Washing Hands, Trimming, Using Hairbrush, Toilet Flushing

Living Room: Exercising, Walking, Eating/Drinking, Vacuuming, Coughing

Getting Activity Labels from A/V is Non-trivial



No 1:1 mapping with daily activities. (e.g., sitting on sofa)

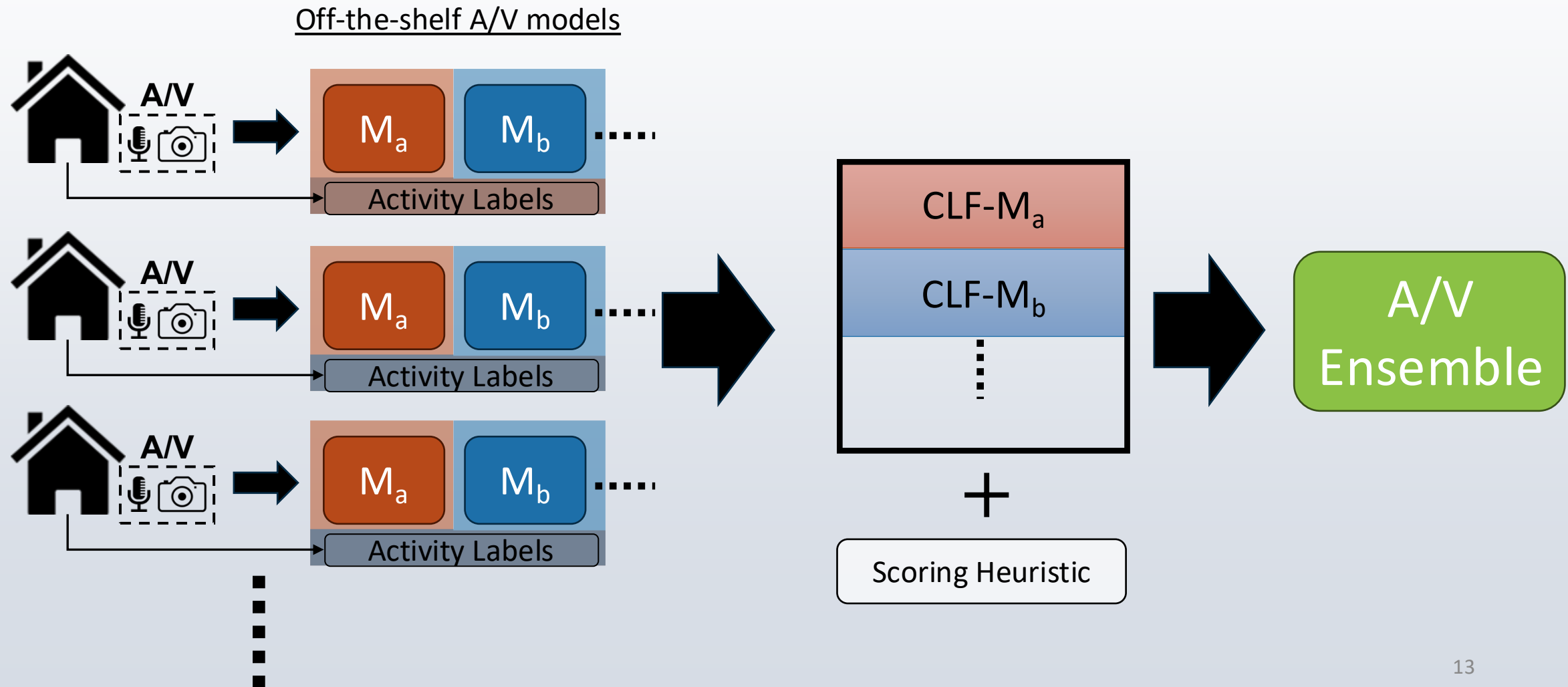


Multiple output predictions from models can have similar signatures. (e.g., motor-based devices)



Different models are suited well for different subset of activities

Learn a Non-Linear Mapping from Predictions to Activity Labels



Needs High Precision for Clean Activity Labels

Activity Detection Rate: Labels predicted with high confidence from A/V Ensemble.

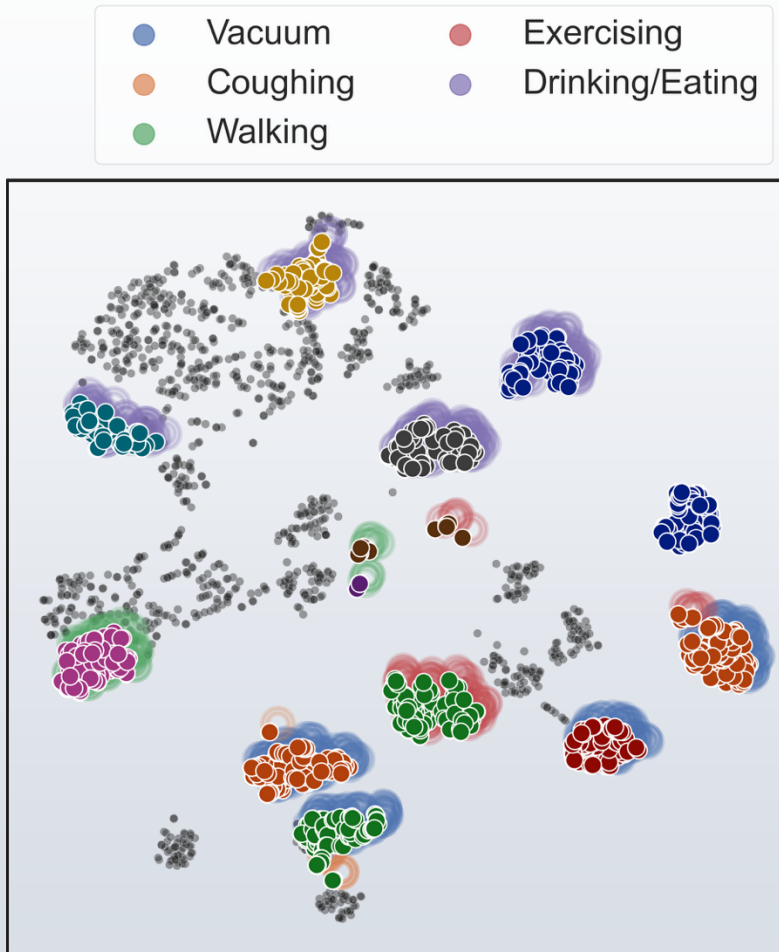


Low (65%) detection rate for high (>90%) precision using A/V ensemble.

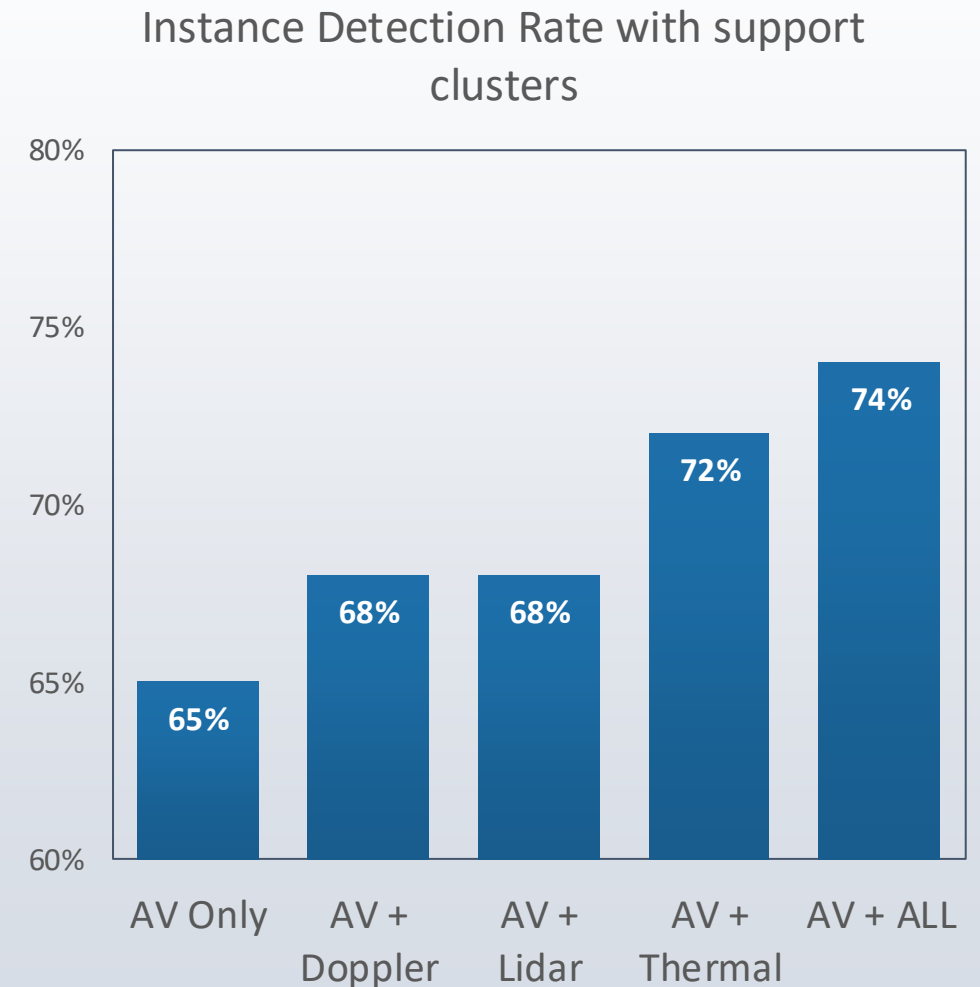


Insight: Utilize information from “X” sensors to increase detection rate.

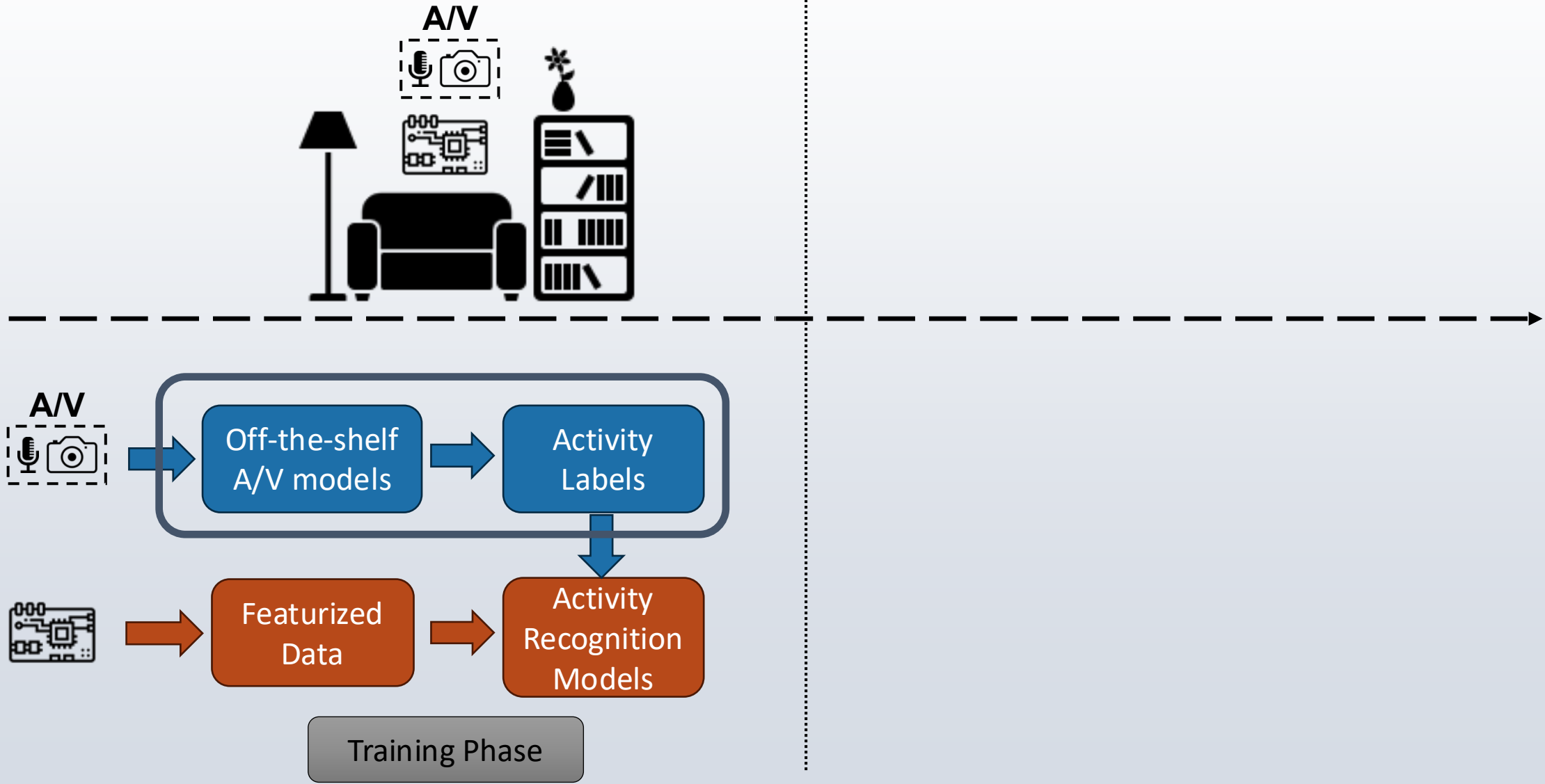
Increasing Activity Detection Rate using “X” Sensors



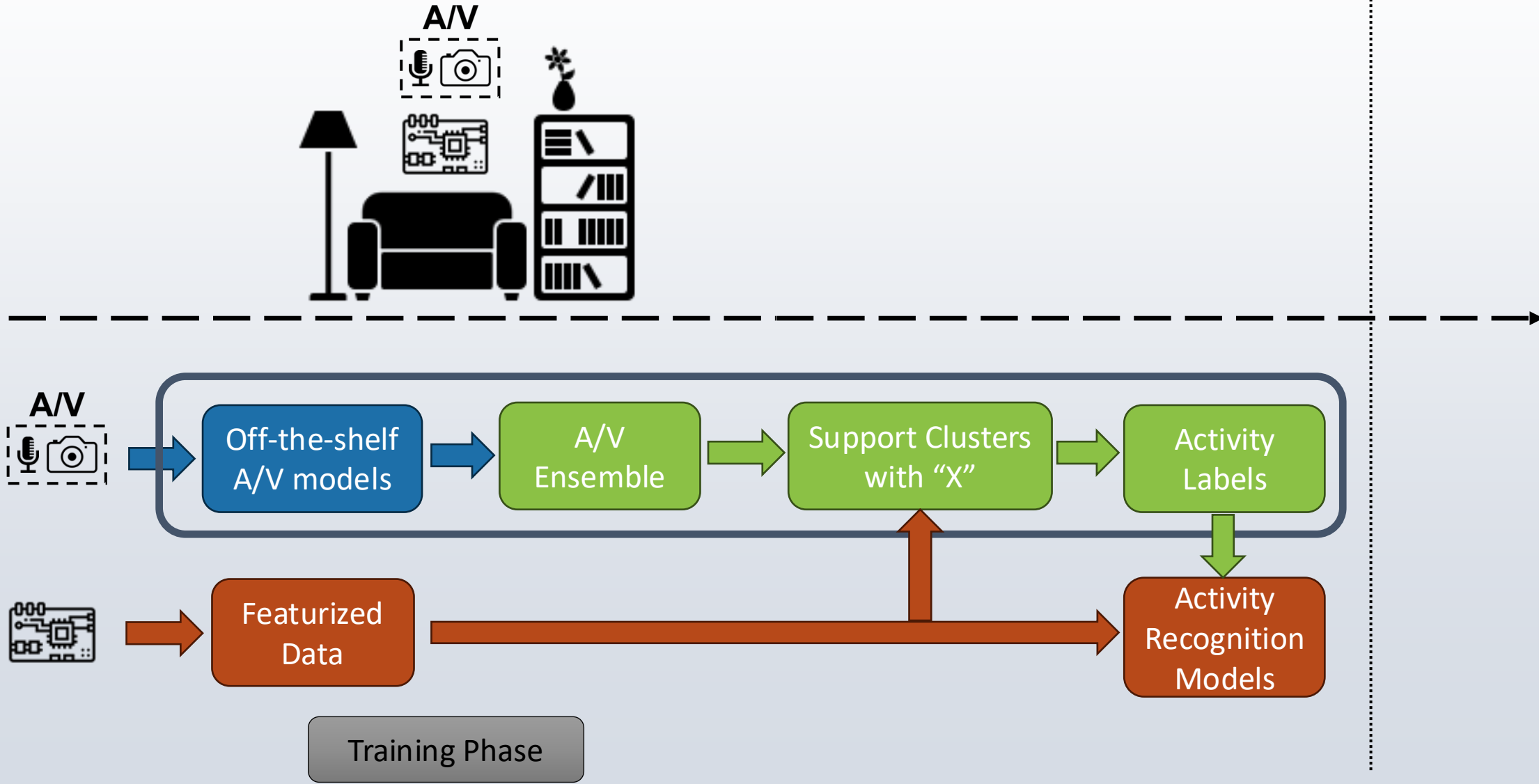
2D visualization for Lidar data



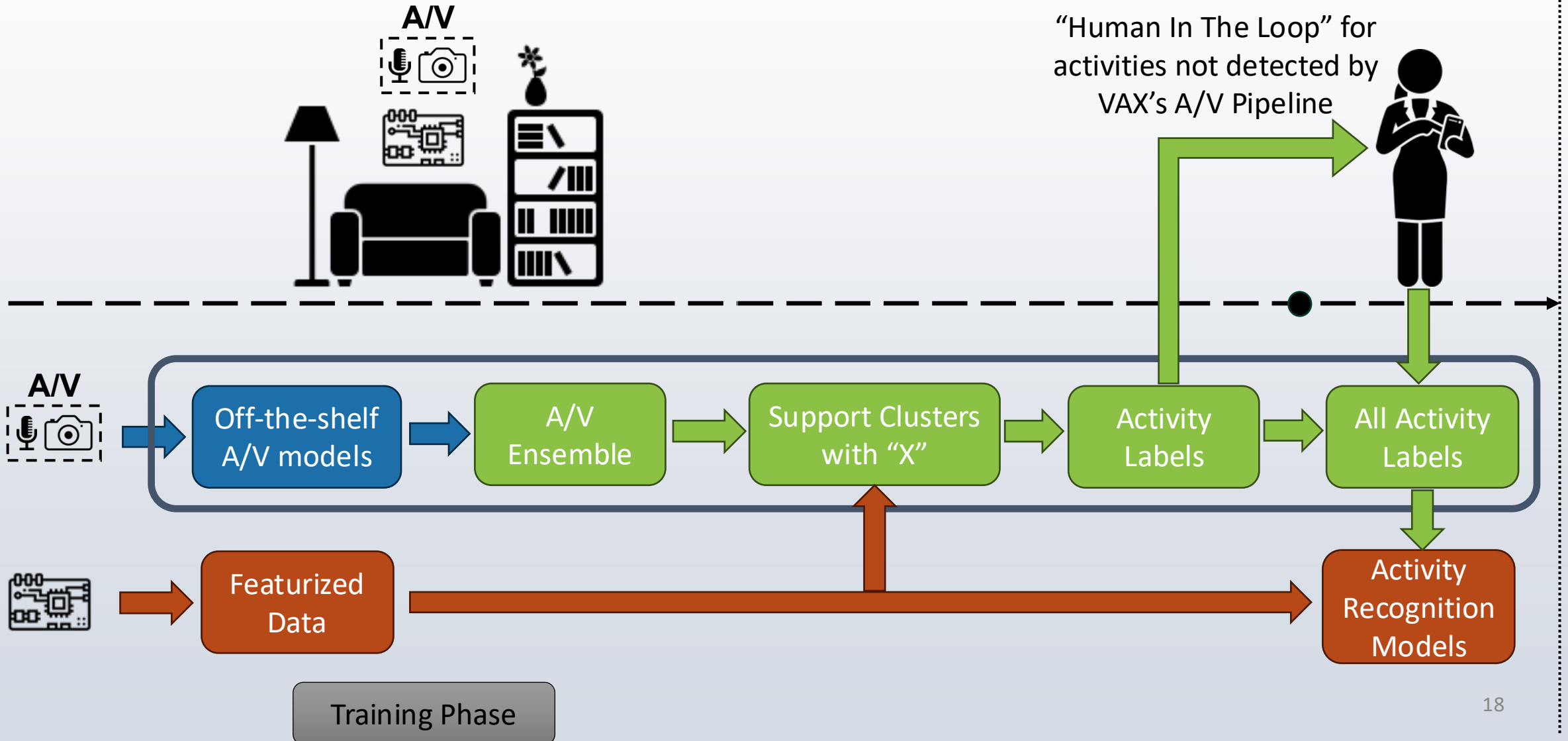
VAX (Video/Audio to “X”)



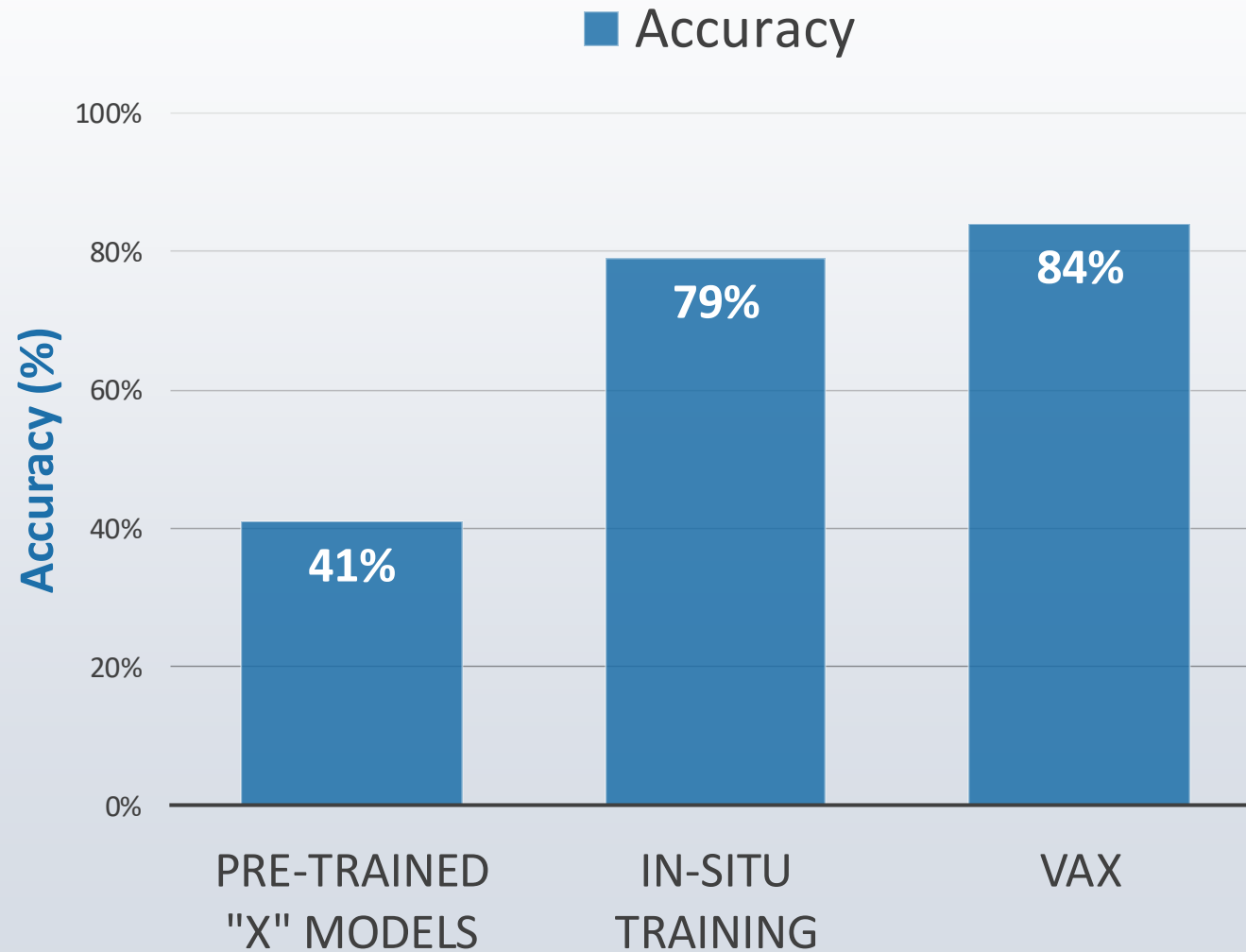
VAX (Video/Audio to “X”)



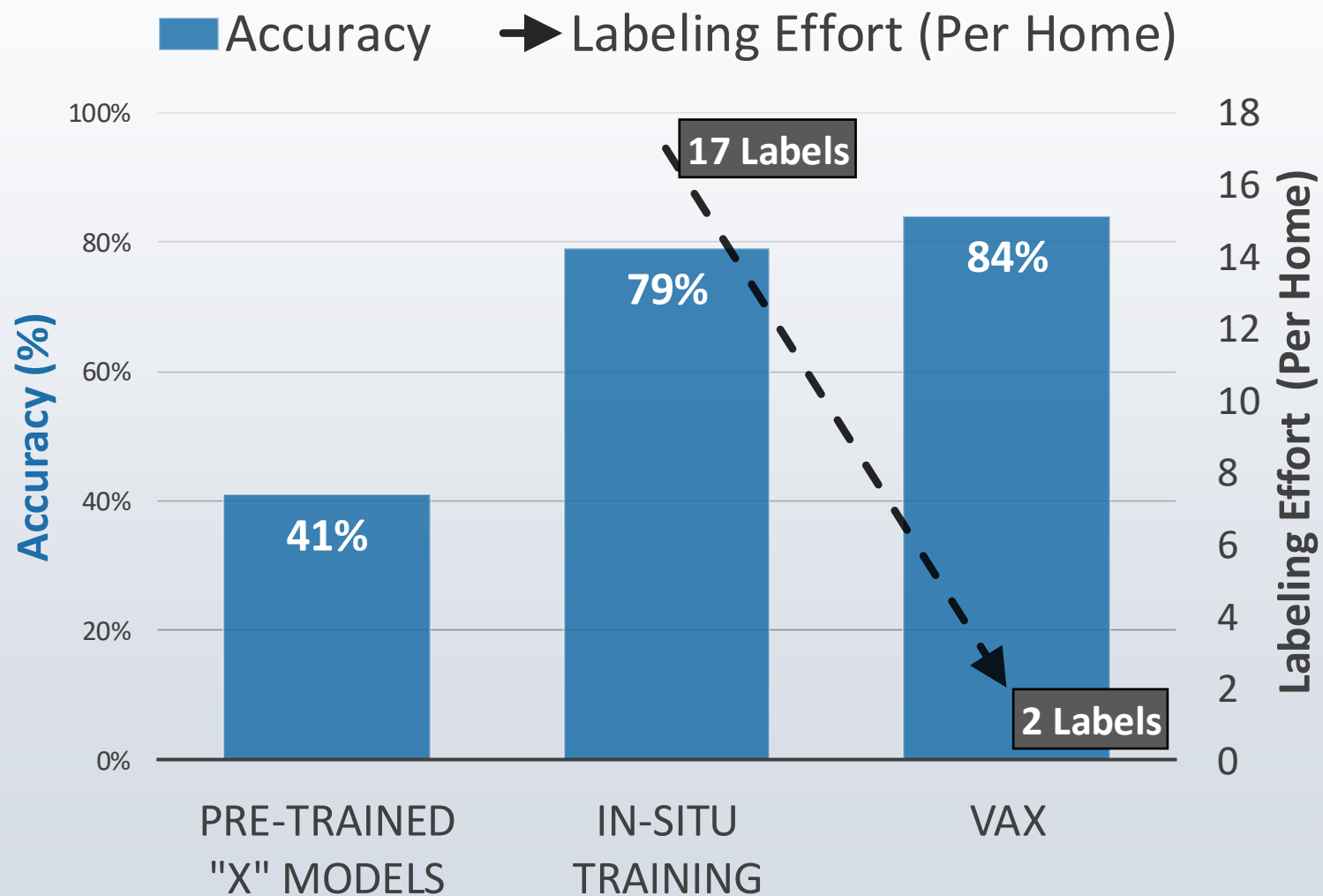
VAX (Video/Audio to “X”)



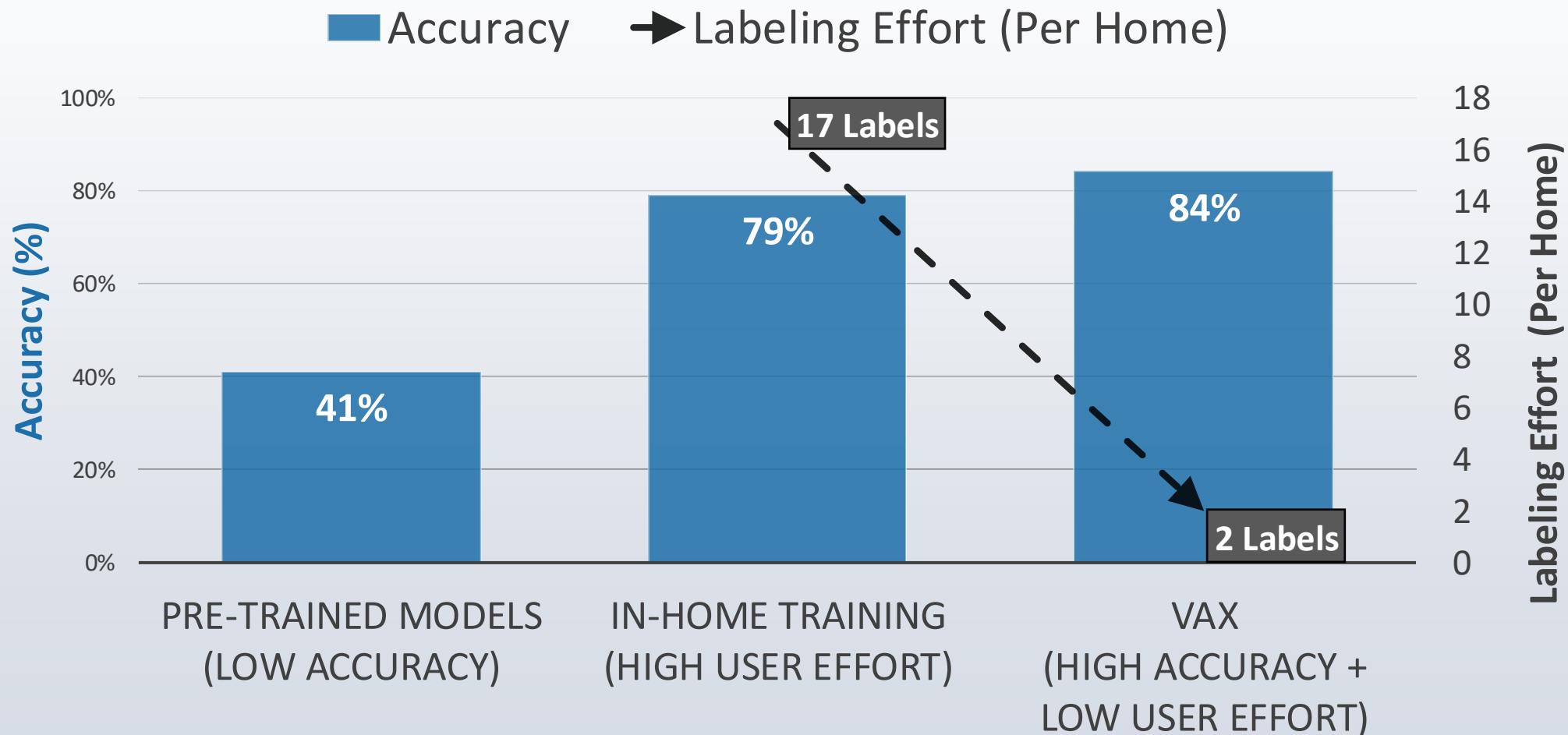
Comparison with Baseline Approaches



Comparison with Baseline Approaches

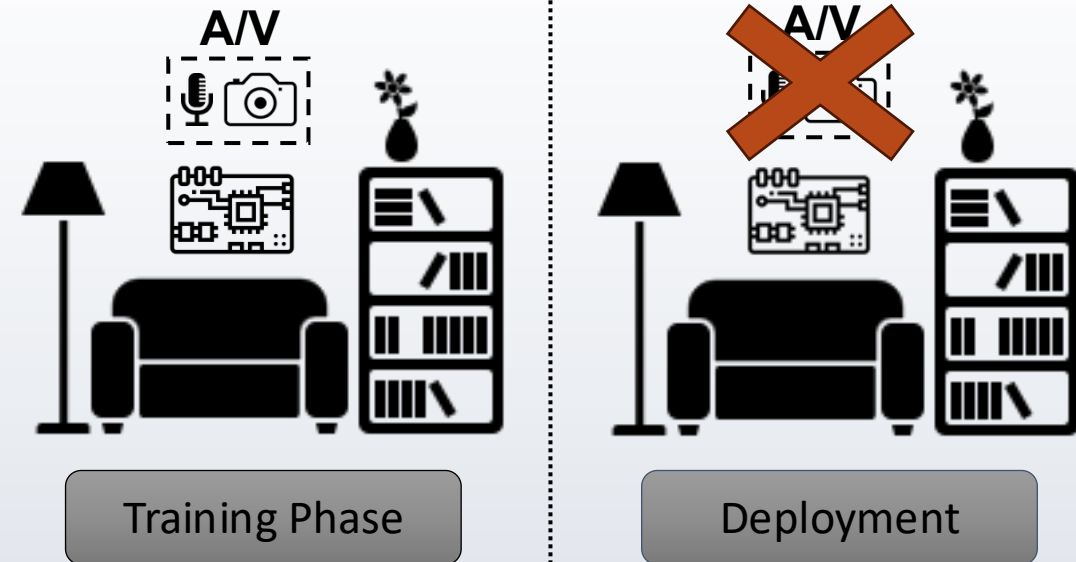


Comparison with Baseline Approaches



Conclusion

An end-to-end system to utilize off-the-shelf A/V models to bootstrap HAR for any “X” sensing modalities.



Prasoon
Patidar



Mayank
Goel



Yuvraj
Agarwal

VAX is open source! Visit the link on QR Code for more details.

github.com/synergylabs/vax

